5.4 Exponential and Logarithmic Equations

Name _____ Period ____ Date _____

Pre-Calculus 5.4 · Student sheet

I can…

- solve exponential equations.
- solve logarithmic equations.
- check for extraneous solutions.

Vocabulary · fill in the missing word

Extraneous solution — a solution of a _____ equation that does not satisfy the original equation

Part 1 · Which tool?

Sort: $2^x = 32$, $e^x = 7$, $\log_3 x = \log_3 12$, $3^x = 20$, $\ln x = 2$, $4^x = 8$

Same base: One-to-One	Not the same base: logs or exponents

Part 2 · Notes and examples (fill in each note, then work the example)

One-to-One Properties: $a^x = a^y \iff x = y$; $\log_a x = \log_a y \iff$ _____.

Example 1

Solve.

a. $2^x = 32$

b. $\ln x - \ln 3 = 0$

c. $\left(\frac{1}{3}\right)^x = 9$

Inverse Properties: to solve $a^x = b$, take the _____ of both sides; to solve $\log_a x = b$, exponentiate: $x = a^b$.

Example 2

Solve. Round to three decimal places.

a. $e^x + 5 = 60$

b. $2(3^{2t-5}) - 4 = 11$

Example 3

Solve $e^{2x} - 3e^{x} + 2 = 0$.

Check every solution in the original equation: logarithms of _____ numbers are undefined.

Example 4

Solve $\log x + \log(x - 9) = 1$.

Example 5

a. Solve $5 + 2 \ln x = 4$.

b. How long does money take to double at 6% compounded continuously?

Part 3 · Partner practice

Partner 1

Solve $3^{x} = 20$. Round to three decimal places.

Partner 2

Solve $\log_2(x + 1) + \log_2(x - 1) = 3$.